

Subject 1 (7+2 minutes)

Due February 20, 2021 1:29 PM • Closes February 20, 2021 1:30 PM

Instructions

- a) For the given graph, apply Prim algorithm starting at the vertex a. Write the candidate edges at each step.
- b) Which problem is solved with this algorithm?
- c) Calculate the weight of the output graph.

Subject 3 (10+2 minutes)

Due February 20, 2021 11:40 AM • Closes February 20, 2021 11:41 AM

Instructions

In the given graph, find a (b,h) -walk of minimum length, which contains each edge of the graph at least once. If the weight of the graph is 95, find the length of this walk.

Subject 5 (5+2 minutes)

Due February 20, 2021 12:17 PM • Closes February 20, 2021 12:18 PM

Instructions

Find the general term of the sequence $a(n)$ given by the recurrence relation $a(n+3)=3a(n+2)-a(n+1)+3a(n)$, with $a(0)=5$, $a(1)=8$, $a(2)=5$.

Subject 6 (12+2 minutes)

Due February 20, 2021 12:35 PM • Closes February 20, 2021 12:36 PM

Instructions

We distribute \$20 to 6 persons. We admit only integer sums to each person. In how many ways can we distribute the money if:

- a) we allow persons who don't receive anything
- b) we allow persons who don't receive anything, but the whole sum cannot be given to only one person
- c) each person receives at least \$1
- d) there are (it is sure there are) persons that don't receive anything (sum with >2 terms not allowed as answer)
- e) there are exactly 2 persons who don't receive anything
- f) exactly one person receives as much as all other persons together.

Explain your answers!

Subject 7 (8+2 minutes)

Due February 20, 2021 12:48 PM • Closes February 20, 2021 12:49 PM

Instructions

!! You may use a calculator (it's indicated)

In screening the sars-cov-2 infection, the probability that a healthy person wrongly gets a positive result is 0.05. The probability that an infected person wrongly gets a negative result is 0.02 and the overall rate of the infection in a population is 1%.

- a) If my test gives a positive result, what is the probability I am actually infected?
- b) In the case the overall rate of the infection is 50%, if my test gives a positive result, what is the probability I am actually infected?
- c) Based on the numerical values of the previous results, which is your opinion about testing randomly a population (e.g. Romania, today)?

Subject 8 (15+2 minutes)

Due February 20, 2021 1:07 PM • Closes February 20, 2021 1:08 PM

Instructions

We paint each of the faces of a die in one of the colors red and blue, such that, when we roll it the probability to have a red face is p . Then we roll the die until we obtain faces of different colors, but not more than 5 times (we stop either after we obtain faces of different colors, or, if this doesn't happen, after 5 rolls). Let X be the total number of rolls.

- a) Which are the possible values of X ?
- b) Find $P(X=1)$, $P(X=2)$ and $P(X=6)$
- c) Write the random variable X and prove that it is well defined.
- d) For $p=1/2$, calculate the expected number of rolls and $\text{var}(X)$

Subject 9 (4+2 minutes)

Due February 20, 2021 1:16 PM • Closes February 20, 2021 1:17 PM

Instructions

For the random variable X in the previous problem, write the idea expressed by the law of large numbers.

My work